About 2,600 results for “review wildfire spread prediction”

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Future projections of Siberian wildfire and aerosol emissions

R. K. Nurrohman | Tomomichi Kato | +6 authors | R. Hirata | Environmental Science ·

Biogeosciences · 26 September 2024

Abstract. **Wildfires** are among the most influential disturbances affecting ecosystem structure and biogeochemical cycles in Siberia. Therefore, accurate fire modeling via dynamic global vegetation… Expand

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Soil Moisture is a Stronger Predictor of Forest Fire Spread Potential Than Weather in the U.S. Northern Rocky Mountains

Zachary A. Holden | Alan K. Swanson | M. Sadegh | C. Luce | Erin Noonan-Wright | Russell A. Parsons

Environmental Science · Geophysical Research Letters · 27 August 2025

Accurate **prediction** of forest fire **spread** is a critical management and scientific challenge as the world adapts to rapidly changing fire regimes. We reconstructed 5,400 daily burned area progression… Expand

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Next-Generation AI for Firefighter Hazard Prediction: Exploring BERT and Doc2Vec to Advance Safety and Risk Intelligence

Swarnamouli Majumdar | Computer Science, Environmental Science ·

2025 IEEE World AI IoT Congress (AIIoT) · 28 May 2025

TLDR Comparative results demonstrate that BERT significantly outperforms Doc2Vec in accuracy, context sensitivity, and interpretability, making it ideal for high-stakes applications, and contributes to the development of robust, real-time predictive models. Expand

IEEE | Save | Cite

Advancements in Artificial Intelligence Applications for Forest Fire Prediction

Hui Liu | Lifu Shu | Xiaodong Liu | Pengle Cheng | Mingyu Wang | Ying Huang

Environmental Science, Computer Science · Forests · 19 April 2025

TLDR This study aims to provide valuable support for forest natural resource management and enhanced environmental protection through the application of remote sensing technologies and artificial intelligence algorithms. Expand

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Deep Learning Methods for Lung Cancer Detection, Classification and Prediction - A Review

Ganga V Saji | Thasneem Vazim | S. Sundar | Medicine, Computer Science ·

International Conference on Management… · 18 November 2021

TLDR Different Deep Learning methods used for the detection, classification and **prediction** of cancerous lung nodules and the identification of their malignancy levels are studied and their advantages and limitations are analyzed. Expand

9 | IEEE | Save | Cite

Terrestrial carbon dynamics in an era of increasing wildfire

Tara W. Hudiburg | Justin M. Mathias | +6 authors | Laurel M. Lynch | Environmental Science ·

Nature Climate Change · 1 December 2023

Fire severity is expected to increase as a result of warming. This will potentially amplify climate change due to its impact on the carbon cycle. This **Review** discusses ecosystem carbon loss and… Expand

23 Springer Nature Save Cite

Fire spread prediction using a lagged weather forecast ensemble

S. Louisa S. Matthewsb Environmental Science · 2015

From October to early December 2014 the Bureau of Meteorology conducted a pilot trial of a Rapid Update Cycle (RUC) numerical weather model. The model was run at a horizontal resolution of… Expand

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Research progress and hotspot evolution analysis of wildfire disasters in national park communities

Guorui Zheng Han Xu Xinya Ling Janwen Dong Environmental Science, Biology · Highlights in Science Engineering and Technology · 8 April 2024

TLDR This study used CiteSpace and Bibliometric R analysis tools to conduct a detailed analysis of 328 kinds of literature in the Web of Science database from 2002 to 2023 and provided a reference for promoting **wildfire** hazard risk identification and management in national park communities. Expand

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Towards predictive data-driven simulations of wildfire spread - Part II: Ensemble Kalman Filter for the state estimation of a front-tracking simulator of wildfire spread

M. Rochoux C. Emery S. Ricci B. Cuenot A. Trouvé Environmental Science · 28 May 2014

TLDR Results indicate that data-driven simulations are capable of correcting inaccurate **predictions** of the fire front location and of subsequently providing an optimized forecast of the **wildfire** behavior at future lead times. Expand

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Considerations on operational wildfire spread modelling

R. McRae Environmental Science · 2011

Various issues relating to improving operational **prediction** of **wildfire spread** are discussed. The information content required by fire managers is limited, and can be used to optimise model design.… Expand

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